

FARI

Example Investigation Report

SpaceCAN Spoofing and Replay

Release source: 04-spacecanbus-spoofing-replay-report.md

FARI v1.1.0 | 2026-06-12

This is the versioned release edition of the FARI documentation bundle. Canonical working sources for this release are stored under the same version folder.

DOES NOT MEET

Qualitative executive conclusion from the example report

Document Control

- **Assessment ID:** ASM-FARI-2026-004
- **Investigation ID:** INV-SCAN-001
- **Report version:** 0.3
- **Report maturity:** Provisional
- **Report date:** 2026-06-08
- **Source material:** cases/04_Spacecanbus.md (../cases/04_Spacecanbus.md) and supplied SpaceCAN screenshots
- **Evidence producer:** Auditor; identity not supplied
- **Report author:** FARI Report Author
- **Mission owner / risk authority:** Not supplied
- **Framework profile:** SPARTA live content accessed 2026-06-08

1. FARI Result Card

Field	Result
Conclusion	Does Not Meet
Confidence	Medium within the lab monitoring scope
Required Action	Remediate, then Retest
Priority	Immediate before SpaceCAN monitor values are trusted for operational or autonomous decisions
Scope Boundary	Supplied SpaceCAN lab injection/replay tools and monitor screenshots; physical subsystem response, flight autonomy, production bus controls, and on-orbit behavior excluded

Rationale: The supplied evidence demonstrates a forged reply-like frame, replay transmission, and changed displayed telemetry values without evidence that the displayed data was authenticated, attributed as untrusted, or rejected. This demonstrates failure of the lab-scope claim. The evidence does not prove physical actuator effects or mission-level consequence.

2. Executive Decision Summary

The SpaceCAN lab monitor accepts or displays bus state that can be influenced by injected reply-like traffic. A before/after monitor screenshot shows thruster values changing from 12 to 255 while status remains nominal. The replay tool also demonstrates retransmission of captured bus frames.

This is sufficient to conclude that the claim fails in the evidenced lab monitoring scope. It is not sufficient to conclude that physical thrusters, flight-control logic, or on-orbit systems accepted or acted on the values.

Decision required: Do not use unauthenticated SpaceCAN monitor data as a trusted basis for operational or autonomous action until origin validation, freshness controls, anomaly handling, and retest evidence are available.

3. Frame

Mission Context Added by the Report Author

The supplied material does not identify a mission architecture or SpaceCAN consumer. This report uses the explicit assumption that the monitored bus values could inform operators, health management, or autonomous logic. That assumption requires mission-owner confirmation.

Scope

Item	FARI Record
Primary asset	AST-SCAN-BUS-001 SpaceCAN lab bus
Related assets	AST-SCAN-MON-001 SpaceCAN monitor and AST-SCAN-NODE-004 represented node 0x04
Segment	Space-segment representative onboard bus lab
Access model	White box, as stated by evidence producer
Environment	Dynamic local lab using supplied SpaceCAN examples
Claim	CLM-SCAN-001: SpaceCAN monitoring and consumers reject or visibly distinguish unauthenticated forged or replayed reply-like frames before treating them as trusted state
Explicit exclusions	Physical subsystem behavior, real spacecraft bus controls, autonomous decision logic, RF/ground paths, and on-orbit behavior

Authorization

No rules-of-engagement reference, target revision, test date, bus topology, safety control, or evidence-handling record was supplied.

4. Acquire: Source-Material Intake Review

Evidence Index

Evidence ID	Source	Directly Supported Statement
EVD-SCAN-001	<code>spacecan_scinjection_cmd_s poof.png</code>	The injection tool sent a frame to ID 0x304 with DLC 2 and data 01 FF
EVD-SCAN-002	<code>spacecan_screenplay_replay .png</code>	The replay tool retransmitted multiple recorded frames, including ID 0x304
EVD-SCAN-003	<code>Spacecan_tui_value_change. png</code>	The monitor comparison shows thruster values changing from 12 to 255 while status remains nominal

Evidence-Backed Facts

- The supplied injection command is `./buildDir/examples/scinjection 0x304 01 FF`.
- The injection screenshot shows a sent frame with identifier 0x304, DLC 2, and data 01 FF.
- The replay screenshot shows replay activity and multiple frame identifiers, including 0x304.
- The monitor comparison shows values labeled for thrusters changing from 12 to 255; the displayed status remains nominal.

Evidence-Producer Assertions

- Frame ID 0x304 represents a reply from node 0x04.
- Monitors or consumers trust reply CAN IDs and attribute the payload to the forged node.
- The replay preserves original inter-frame timing.
- No sequence counter, nonce, timestamp authentication, or freshness window blocks the replay.
- Forged or replayed traffic can mask subsystem failure or poison higher-level decision logic.

Report-Author Inferences

- The lab monitor displays state that can be influenced by injected bus traffic without an evidenced untrusted-data indication.
- The displayed value change is consistent with the supplied injection scenario, but the exact chronology and causal capture are not fully documented.

- Replay transmission is demonstrated; receiver acceptance and a replay-specific state change are not separately isolated.

Contradictions and Quality Issues

ID	Issue	Reporting Decision
GAP-SCAN-001	The final finding says the library has DoS in APID 0x06, which is unrelated to SpaceCAN evidence	Exclude the copied finding statement
GAP-SCAN-002	The same monitor screenshot is used for spoofing and replay	Confirm displayed-state manipulation in the lab, but keep the replay-specific effect candidate
GAP-SCAN-003	The exact application meaning of payload 01 FF is not evidenced	Describe the observed display change without claiming a universal field encoding
GAP-SCAN-004	No source/configuration evidence demonstrates absence of authentication or freshness controls	Treat missing controls as an evidence request, not an observed fact
GAP-SCAN-005	Status remains nominal while values change, but alarm semantics are not supplied	Record a possible detection gap without confirming it

Sufficiency Decision

Question	Decision	Rationale
Technical condition	Sufficient for lab display-trust failure; partial for replay-specific acceptance	Injection/replay actions and displayed value change are evidenced, but replay causality is not isolated
External / operational reachability	Partial	Local bus access is demonstrated; production/on-orbit access paths are unknown
Mission consequence	Partial	Misleading displayed state is observed; operator, autonomy, and physical effects are not demonstrated

5. Relate: Findings and Mission Context

FND-SCAN-001: Forged Reply-Like Frame Influences Displayed Telemetry

- **State:** Confirmed within the lab monitor scope
- **Condition:** Reply-like SpaceCAN traffic can be injected using a valid bus identifier and displayed without an evidenced untrusted-origin distinction.
- **Observed effect:** The supplied monitor comparison shows displayed thruster values changing from 12 to 255 while status remains nominal.
- **Preconditions:** Local bus injection access and a consumer that processes the forged identifier and payload.
- **Credible consequence:** Operator or automated logic may use misleading state if equivalent behavior exists in the operational system.
- **Technical confidence:** Medium
- **Operational-reachability confidence:** Low
- **Mission-consequence confidence:** Low

FND-SCAN-002: Captured SpaceCAN Traffic Can Be Replayed

- **State:** Candidate for receiver/control failure; replay transmission confirmed
- **Condition:** The supplied replay tool retransmits captured bus frames, including ID 0x304.
- **Observed effect:** Replay transmission is shown; a receiver reaction caused specifically by replay is not isolated.
- **Credible consequence:** Stale telemetry or commands may be accepted again if operational consumers lack freshness controls.
- **Technical confidence:** Medium for transmission; low for receiver effect.

Framework Mappings

Mapping	State	Rationale
SPARTA EX-0014.02 Bus Traffic Spoofing (https://sparta.aerospace.org/technique/EX-0014/02/)	Confirmed in lab scope	The injection uses an internal-bus identifier to make forged data appear to originate from a represented node
SPARTA EX-0001.02 Bus Traffic Replay (https://sparta.aerospace.org/technique/EX-0001/02/)	Confirmed for transmission; candidate for receiver effect	The replay tool retransmits previously captured internal-bus frames

Mapping	State	Rationale
SPARTA DE-0002.03 Inhibit Spacecraft Functionality (https://sparta.aerospace.org/technique/DE-0002/03/)	Candidate	Applicable only if misleading or stale values inhibit spacecraft functionality; that effect is not demonstrated
SPARTA REC-0005.01 Uplink Intercept Eavesdropping (https://sparta.aerospace.org/technique/REC-0005/01/)	Not selected	The evidence shows a local replay file, not interception of an uplink

Qualitative Mission Context

Given local access to the SpaceCAN bus, an actor or faulty node could inject or replay reply-like frames that appear as trusted telemetry, potentially misleading operators or automated consumers and harming mission decisions.

The displayed-state integrity failure is demonstrated only in the lab scope. Physical and mission consequences remain provisional.

6. Inform

Required Actions

1. Define which SpaceCAN messages require origin authentication, integrity, and freshness protection.
2. Reject or visibly label frames that fail origin, sequence, timing, mode, or plausibility checks.
3. Correlate critical telemetry with independent sources and expected state transitions.
4. Alert when values change implausibly or conflict with status.
5. Separate monitoring-only data from values allowed to drive control or fault management.
6. Retest spoofing and replay against every operational consumer, not only the lab display.

Acceptance Criteria

- Forged reply-like frames are rejected or clearly marked untrusted.

- Previously accepted frames cannot be replayed outside the defined freshness window without detection and rejection.
- Critical consumers do not act on unauthenticated or stale state.
- Alerts distinguish inconsistent status and telemetry values.
- Retest evidence covers monitor, control, autonomy, and recovery behavior in a representative integration.

Evidence Gaps and Follow-Up Requests

Gap ID	Requested Evidence	Why It Matters
GAP-SCAN-006	Timestamped packet capture correlated with the monitor change	Establish exact causality
GAP-SCAN-007	Bus topology, node ownership, and trust model	Evaluate operational reachability
GAP-SCAN-008	Receiver/monitor source and security-control configuration	Verify origin and freshness controls
GAP-SCAN-009	Replay-specific before/after capture	Resolve receiver response to replay
GAP-SCAN-010	Physical subsystem and autonomous-consumer test results	Evaluate mission consequence
GAP-SCAN-011	Mission thresholds, alarm semantics, and unacceptable outcomes	Support final mission decision

Assurance Statement

CLM-SCAN-001 **Does Not Meet** its acceptance criteria with medium confidence in the supplied lab monitoring scope. Forged bus traffic and changed displayed telemetry are evidenced, and replay transmission is shown. No conclusion is made about physical subsystem behavior, autonomous action, production controls, or on-orbit consequence.

7. Coverage

Area	Coverage	Result
Local bus injection	Tested and evidenced	Forged frame transmission shown
Lab monitor displayed values	Tested and evidenced	Displayed telemetry changes shown
Bus replay transmission	Tested and evidenced	Recorded frames retransmitted
Replay-specific receiver effect	Inferred / partial	Not isolated

Area	Coverage	Result
Physical subsystem and autonomy	Not assessed	No conclusion
Production and on-orbit environment	Out of scope	No conclusion

8. Closing FARI Result Card

Field	Result
Conclusion	Does Not Meet
Confidence	Medium within the lab monitoring scope
Required Action	Remediate, then Retest
Priority	Immediate before displayed SpaceCAN state is trusted for operational decisions
Next Decision	Approve trusted use only after origin/freshness controls and representative retest satisfy acceptance criteria